

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 - ANALYTICAL PROBLEM SOLVING

Course Code:	CSA0305	Course Title:	Data Structures
CO Assessed:	CO2 - Manipulation (BL3, BL4)	Assessment:	Assessment Tool 1 - Analytical Problem Solving
Weightage:	30% of CIA	Total Marks:	100
CO2 Statement:	Implement linear data structures such as stacks, queues, and linked lists, and analyze the efficiency of linear and binary search techniques.		
Student Name:	K. Karthik	Reg. No.:	19252 5040
Section / Batch:	AI & ML	Date:	20/07/2026

Code of Conduct

I K. Karthik (19252 5040) (Name / Reg No)

certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Karthik

Instructions

- This test contains 5 Analytical problem-solving questions. Total: 100 Marks.
- When a student answer using an Analytical problem-solving, in evaluation the answer should be with following five requirements: Understanding of problem, select appropriate data structure, Design the solution, analyze, Clarity of representation.
- Each student should write 2 questions. No negative marking. Unanswered questions carry zero marks.
- No DTP printouts or electronic devices permitted. They should provide written materials.

Section / Topic	Focus	Q Nos	Marks
Linked Data Structure, Search Data Structure	List Operations, Binary Search	1	50
Stack Data Structure, Queue Data Structures	Stack Notations, Priority Queue	2	50
GRAND TOTAL:			100 Marks

			2. Mr. XYZ booked a ticket on Pandian Express. He reached the station and was looking for his PNR number in the chart. Call out binary search procedure to search the PNR number 4056786. The ticket numbers are 4056779, 4056780, 4056781, 4056782, 4056783, 4056784, 4056785, 4056786, 4056787, 4056788, 4056789.
12	192472154	KARNATI VIGNESH	1. Convert the expression $((a + b) + c * (d + e) + f) * (g + h)$ to 1) Prefix expression, 2) Postfix expression. 2. Two students obtain different postfix expressions for the same infix expression. Infix: $(A+B)*(C-D)/E$ Student A: $AB+CD-*E/$ Student B: $AB+C*D-E/$ Determine which postfix expression is correct and justify your answer by tracing the conversion process.
13	192521157	KABILAN K	1. How a queue may be maintained by a circular array with $MAXSIZE = 6$. Observe that queue always occupies consecutive locations except when it occupies locations at the beginning and at the end of the array. If the queue is viewed as a circular array, this means that it still occupies consecutive locations. The queue is empty when $count = 0$; Perform the following Enqueue and Dequeue operations. a) Initially QUEUE is empty, b) A, B, C are Enqueued, c) A is Dequeued, d) D, E, F are Enqueued, e) B & C are Dequeued, f) G is Enqueued, g) D & E are Dequeued, h) H & I are Enqueued, i) F is dequeued, j) G & H are dequeued k) I is Dequeued. Demonstrate with a neat sketch of Front, Rear and count in a queue and circular queue. 2. Show the linked list configuration resulting from each series of list operations using the List ADT. Assume that lists L1 and L2 are empty at the beginning of each series. Show where the current position is in the list. (a) $L1.append(10); L1.append(20); L1.append(15);$ (b) $L2.append(10); L2.append(20); L2.append(15); L2.moveToStart(); L2.insert(39); L2.next(); L2.insert(12);$
14	192572105	KATTAMURI VENKATA SRINIVASAGNANES WAR KUMAR	1. Evaluate the following postfix expression using a stack. Show the stack contents after every operation. $23*54*+9-$ 2. Design an algorithm to evaluate any postfix expression using a stack. Explain why stacks are more suitable than queues for postfix evaluation.
15	192525040	KONIJETI KARTHIK	1. Given a singly linked list L: $L1 \rightarrow L2 \rightarrow L3 \dots \rightarrow L_{n-1} \rightarrow L_n$, reorder and print it to: $L1 \rightarrow L_n \rightarrow L2 \rightarrow L_{n-1}$. 2. Insert the following elements 150, 250, 350, 450, 550 into the double linked list and delete the elements 150 and 250 from the double linked list. Show the linked structure using a node diagram before and after delete operations.

38	192524251	VAISHNAV M	- Two algorithms solve the same problem. Algorithm A uses recursion, while Algorithm B uses an explicit stack. Compare both approaches in terms of memory usage, execution flow, and risk of stack overflow. - A stack has a capacity of 5 elements. Explain what happens if the following operations are executed: Push(A), Push(B), Push(C), Push(D), Push(E), Push(F). Identify the error and suggest two possible solutions.
39	192525187	VEERA SANTHOSH A	- Compare the performance of a linear queue and a circular queue when 100 enqueue and dequeue operations are performed alternately. - A queue implemented using an array report "Queue Full" even though some positions are empty. - Analyze why this occurs. - Explain how a circular queue solves this problem.
40	192511226	VINIL KUMAR S	- Consider the singly linked list: $10 \rightarrow 20 \rightarrow 30 \rightarrow 40 \rightarrow 50$. Delete the node containing 30. Explain how the links change after deletion. - Given two sorted linked lists: L1: $1 \rightarrow 3 \rightarrow 5 \rightarrow 7$ L2: $2 \rightarrow 4 \rightarrow 6 \rightarrow 8$

Rubrics for Calculation

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Problem Understanding	20	Demonstrates complete and precise understanding; identifies all constraints and edge cases	Shows clear understanding with minor gaps	Partial understanding; misses some constraints	Misinterprets problem or lacks clarity
Data Structure Selection	20	Selects optimal data structure with strong justification and comparison	Selects appropriate structure with reasonable justification	Selects somewhat suitable structure with weak reasoning	Incorrect or no data structure selection
Algorithm Design	20	Designs efficient, logically sound, and well-structured algorithm	Algorithm is correct with minor inefficiencies	Algorithm is partially correct or lacks clarity	Algorithm is incorrect or missing
Accuracy of Solution	30	Error-free, well-structured, optimized code/solution	Minor errors but overall functional	Multiple errors; partially working	Non-functional or missing solution
Clarity	10	Exceptionally clear, well-organized, uses diagrams effectively	Clear and organized with minor issues	Somewhat organized but lacks clarity	Poorly organized and unclear

Faculty In-charge	Course Coordinator	Program Director
Signature: <u>[Signature]</u>	Signature: _____	Signature: _____
Date: <u>20/7</u>	Date: _____	Date: _____

a singly linked list $L: L_1 \rightarrow L_2 \rightarrow L_3 \dots \rightarrow L_{n-1} \rightarrow L_n$ reorder and print

singly Linked List Reordering

Given:
 $L_1 \rightarrow L_2 \rightarrow L_3 \rightarrow L_4 \rightarrow L_5 \rightarrow \dots \rightarrow L_n$

reordered list:

$L_1 \rightarrow L_n \rightarrow L_2 \rightarrow L_{n-1} \rightarrow L_3 \rightarrow L_{n-2} \rightarrow \dots$

for example:

Before:

$L_1 \rightarrow L_2 \rightarrow L_3 \rightarrow L_4 \rightarrow L_5$

After:

$L_1 \rightarrow L_5 \rightarrow L_2 \rightarrow L_4 \rightarrow L_3$

2. Doubly Linked List

elements inserted:

Explanation:

The linked list has to be re-arranged by taking nodes alternately from the beginning and from the end.

The first node L_1 remains in the first position. Then the last node L_n is placed after it. Next, L_2 is taken from the beginning, followed by L_{n-1} from the end.

the linked list:

$1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 5 \rightarrow 6$

The first node is 1 and the last node is 6. The first two nodes of the new list are:

$1 \rightarrow 6$

Next, take the second node 2 and the second last

5:

$1 \rightarrow 6 \rightarrow 2 \rightarrow 5$

Then take 3 and 4:

$1 \rightarrow 6 \rightarrow 2 \rightarrow 5 \rightarrow 3 \rightarrow 4$

Therefore, the final re-ordered linked list is:

original list:

$1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 5 \rightarrow 6$

Re-ordered list:

$1 \rightarrow 6 \rightarrow 2 \rightarrow 5 \rightarrow 3 \rightarrow 4$

Steps involved:

1. Start from the first node of the linked list.
2. Identify the last node.
3. Place the last node after the first node.
4. Move to the second node from the beginning.
5. Find the second-last node and place it next.
6. Continue this process alternately from the beginning and at the end.
7. Stop when all nodes are included in the re-ordered list.

Example:-

For:

$L1 \rightarrow L2 \rightarrow L3 \rightarrow L4 \rightarrow L5$

reordered sequence is:

$L1 \rightarrow L5 \rightarrow L2 \rightarrow L4 \rightarrow L3$

Same procedure is followed for any number of nodes.

the following elements 150, 250, 350, 450, 550 into the double linked list and delete the elements 150 and 250 from the double linked list. show the linked structure using a node diagram.

The following elements are inserted into a doubly linked list.

150, 250, 350, 450, 550

A doubly linked list contains three parts in every node:

+ previous pointer (prev)

* Data

+ Next pointer (next)

previous pointer points to the preceding node,

while the next pointer points to the following node.

Step 1: Insert 150

Initially, the list is empty. When 150 is inserted, it becomes the first node.

$\text{NULL} \leftarrow [150] \rightarrow \text{NULL}$

Since it is the first node:

prev = NULL

and next = NULL

Step 2: Insert 250

The new node 250 is inserted after 150

$\text{NULL} \leftarrow [150] \rightleftarrows [250] \rightarrow \text{NULL}$

Here:

* 150's next points to 250

* 250's previous points to 150.

* 250's next is NULL

Step 3: Insert 350

The node 350 is added at the end.

$NULL \leftarrow [150] \rightleftarrows [250] \rightleftarrows [350] \rightarrow NULL$

Step 4: Insert 450

The node 450 is inserted after 350.

$NULL \leftarrow [150] \rightleftarrows [250] \rightleftarrows [350] \rightleftarrows [450] \rightarrow NULL$

Step 5: Insert 550

Finally, 550 is inserted after 450.

Therefore, the complete doubly linked list before

deletion is:

$NULL \leftarrow [150] \rightleftarrows [250] \rightleftarrows [350] \rightleftarrows [450] \rightleftarrows [550] \rightarrow$
 $NULL$

Deletion of 150

Since 150 is the first node, the head is moved from 150 to 250.

After deleting 150:

$NULL \leftarrow [250] \rightleftarrows [350] \rightleftarrows [450] \rightleftarrows [550] \rightarrow NULL$

Final Answer:

Before:

$NULL \leftarrow 150 \rightleftarrows 250 \rightleftarrows 350 \rightleftarrows 450 \rightleftarrows 550 \rightarrow NULL$

Delete:

150 and 250

After:

$NULL \leftarrow 350 \rightleftarrows 450 \rightleftarrows 550 \rightarrow NULL$